

BENNETT UNIVERSITY

School of Computer Science Engineering & Technology

MILESTONE 01

AI IN HEALTHCARE

Problem Domain Identification & Analysis

MediGenius — AI-Powered Medical Information Assistant

CSET343 | AI in Healthcare Mini Project

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1. Domain Context and Problem

Healthcare questions are increasingly asked through search engines and conversational AI. While these tools make information accessible, a fluent answer can still be incomplete, outdated, or unsupported. This is particularly consequential when a user asks about symptoms, medicine use, or an urgent condition: an inaccurate response may delay appropriate professional care.

The project addresses the gap between convenient conversational access and the need for trustworthy, context-aware, safety-conscious health information. Its purpose is to help users understand medical concepts and prepare better questions for clinicians, not to provide autonomous diagnosis or prescriptions.

2. Problem Statement

To design a conversational medical information assistant that interprets natural-language health queries, retrieves relevant knowledge, retains useful conversation context, and screens high-risk requests before presenting understandable, evidence-oriented responses with appropriate referrals and limitations.

3. Analysis of Existing Approaches

Approach	Strength	Limitation motivating this project
General web search	Broad access to health resources	User must judge source credibility and reconcile conflicting information.
General-purpose LLM chat	Natural, interactive explanations	Can produce convincing unsupported statements and unsafe medication guidance.
Static medical FAQ / rule-based bot	Predictable responses to defined prompts	Weak handling of follow-up context and varied user phrasing.
Retrieval-based assistant	Can ground responses in selected documents	Retrieval alone does not prevent unsupported medical conclusions.

4. Proposed Solution: MediGenius

MediGenius combines retrieval-augmented generation (RAG) with controlled agent-based workflows. A safety check first examines whether a request needs refusal or urgent referral. For eligible questions, the system can use a curated medical PDF, selected external information sources, and conversational context to construct an answer. A verification stage checks for unsupported medical claims before the response is returned.

The repository describes LangGraph for agent orchestration, ChromaDB with HuggingFace embeddings for retrieval, Groq-hosted language models, SQLite for conversation history, and a FastAPI–React application layer. These are architectural choices, not evidence of clinical effectiveness.

5. Intended Users and Use Cases

- Students and general users: understand conditions, terminology, and basic health information in plain language.
- Patients before a consultation: organize symptom descriptions and follow-up questions for a clinician.
- Medication information seekers: obtain general explanations while directing dosing or interaction concerns to a pharmacist or doctor.

6. Objectives and Requirements

Area	Milestone-defined requirement
Grounded information	Retrieve relevant passages and distinguish sourced content from uncertain statements.
Conversational usability	Understand follow-up questions within the same user session.
Safety-first behavior	Identify emergency/crisis language; decline unsafe diagnosis, dosing, or interaction requests.
Traceability	Make the basis of retrieved information inspectable; mark uncertainty and include a disclaimer.
System quality	Maintain reasonable response times, protect request data, and handle missing sources/API failures.

7. Scope and Explicit Exclusions

In scope: general health education, explanation of medical terms, grounded Q&A, symptom-description structuring, contextual conversation, and referrals to qualified professionals where needed.

Out of scope: confirmed diagnoses, prescriptions or dose recommendations, definitive drug-interaction clearance, emergency triage as a certified medical device, and replacement of clinician judgement. A crisis keyword gate is a safety measure, not a validated medical risk-assessment system.

8. Risks and Evaluation Plan

Risk	Proposed evaluation / mitigation
Hallucinated clinical claim	Create a test set of questions with known reference answers; manually assess evidence support and unsupported assertions.
Unsafe high-risk response	Test emergency, pediatric/pregnancy dosing, and drug-interaction prompts; record unsafe-compliance rate and referral behavior.
Poor retrieval quality	Evaluate answer relevance and citation grounding against curated medical text; inspect failed queries.
Context or privacy failure	Check session isolation, retention boundaries, logging fields, and behavior with ambiguous follow-ups.
Latency / service failure	Measure per-query response latency over repeated runs and test fallback paths.

Evaluation must distinguish software test coverage from clinical safety. Passing unit tests or showing low latency does not establish medical accuracy or suitability for real-world clinical deployment.

9. Milestone 1 Outcome

The team has identified the healthcare information-access problem, compared baseline approaches, defined MediGenius as a safety-aware information assistant, outlined key requirements and exclusions, and established concrete criteria for subsequent implementation and testing.

References

1. World Health Organization (2021). Ethics and governance of artificial intelligence for health: WHO guidance.
2. Lewis et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS.